Advanced Techniques for EEG Processing - Laboratory Exercises

Exercise 2

Generate three independent signals $s_1(t)$, $s_2(t)$, $s_3(t)$ with duration of 10 seconds and sampled at frequency $F_s = 500\ Hz$. Signal s_1 is a sinusoidal wave with amplitude 0.75 and frequency $5\ Hz$ (use function `sin`). Signal s_2 is a triangle wave having amplitude 1 and frequency $7\ Hz$ (use function `sawtooth`, setting input argument `width` at value 0.5). Signal s_3 is a square wave having 0.5 amplitude and $3\ Hz$ frequency, with a duty cycle of 40% (use function `square`, setting input argument `duty` at 40). Center the previous signals by removing the mean. From the centered signals $S = \begin{bmatrix} s_1 \\ s_2 \\ s_3 \end{bmatrix}$, generate three variables $x_i(t)$, which form the set X, through the following mixing matrix A

$$X = AS \qquad A = \begin{bmatrix} 0.5 & 0.5 & 0.7 \\ 0.7 & 0.2 & 0.4 \\ 0.2 & 0.7 & -0.5 \end{bmatrix}$$

Note that in this case, the number r of the observed variables (X) is equal to the number n of the independent sources ($r = n$).

1) In one figure, plot the three independent components s_1, s_2 e s_3 as a function of time (use `subplot`) and in another figure plot the mixed variables x_i ($i = 1,2,3$) as a function of time (use `subplot`).

2) Compute the principal components of X, using the eigenvectors and eigenvalues decomposition (use `eig` function) and plot the obtained principal components as a function of time (use `subplot`).

3) Is PCA able to recover the original independent components?

4) Obtain the demixing matrix for independent component estimation using the toolbox EEGLAB, through the following steps:

4.a) Save the variables X in a .mat file (function `save`)

4.b) Launch EEGLAB from the Command Window `>> eeglab`

4.c) Import the saved .mat file in EEGLAB. To this aim, from the EEGLAB GUI select '*File>Import Data>Using EEGLAB Functions and Plugins>From ASCII/float file or Matlab array*', then, from the 'Browse' button on the top select the .mat file and fill the value of the sampling frequency ('*Data sampling rate (Hz)*'). Leave unfilled the other edit boxes. Press Ok and assign a name to the dataset. It is possible to visualize the imported signals selecting '*Plot>Channel data (Scroll)*'.

4.d) Estimate the demixing matrix and the independent components of the imported variables by selecting '*Tools>Decompose data by ICA*' and setting

ICA algorithm to use → *Infomax runica.m (default)*

Commandline options → 'extended',1, 'bias', 'off', 'rndreset', 'no'

Check *Reorder components by variance* (should already be checked by default)

Press Ok and wait for the algorithm to terminate. It is possible to visualize the estimated independent components selecting '*Plot>Component activations (Scroll)*'.

4.e) Export the demixing matrix selecting '*File>Export>Weight matrix to text file*'; and give a name to the file, specifying the extension.txt.

4.f) Save the current dataset created in EEGLAB selecting '*File>Save current dataset(s)*', assigning a name (two files are automatically saved with extension .set and .fdt).

4.g) Close the EEGLAB windows. Return in your .m script fileand use function `load` to import the demixing matrix from the saved .txt file.

5) Using the demixing matrix, compute the estimated independent components and plot them as a function of time (use `subplot`).

6) Now from the centered signals $S = \begin{bmatrix} s_1 \\ s_2 \\ s_3 \end{bmatrix}$, generate five variables $x_i(t)$, which form the set X, through the following mixing matrix A

$$X = AS \qquad A = \begin{bmatrix} 0.5 & 0.5 & 0.7 \\ 0.7 & 0.2 & 0.4 \\ 0.2 & 0.7 & -0.5 \\ -0.6 & 0.3 & 0.2 \\ 0.1 & -0.5 & 0.4 \end{bmatrix}$$

Note that in this case, the number r of the observed variables (X) is greater than the number n of the independent sources ($r > n$), and PCA needs to be applied before ICA to reduce to the case $r = n$ (basic ICA model).

7) Plot the mixed variables x_i ($i = 1,2,\dots,5$) as a function of time (use `subplot`). Then, compute the principal components of X, using the eigenvectors and eigenvalues decomposition (use `eig` function) and plot the obtained principal components as a function of time (use `subplot)`.

8) Is PCA able to recover the original independent components? What is PCA able to do?

9) Consider only the first three principal components (having non-zero variance); this corresponds to reduce r equal to n. Then, use them to estimate the independent components by applying the following steps

9.a) Save the first three principal components in a .mat file (function `save`)

9.b) – 9.g) Same steps as 4.b) to 4.g)

10) Using the demixing matrix obtained at previous point 9), compute the estimated independent components and plot them as a function of time (use `subplot`).

Note that the same results as in steps from 7) to 10) could be obtained by importing the entire set of five variables X in eeglab and by applying the algorithm runica with the following Commandline options

Commandline options → 'extended',1, 'bias', 'off', 'rndreset', 'no','PCA', 3.

In this way, the algorithm performs internally the decomposition in PC, takes the first three PCs and estimate on them the demixing matrix.